

ETLs, EDAs, DATA CURATION

ETLs, EDAs, Data Curation, and Visual Storytelling

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FRANCISCO JOSÉ DE CALDAS

- 1 ETL and Exploratory Data Analysis
- 2 Data Curation



Outline

1 ETL and Exploratory Data Analysis

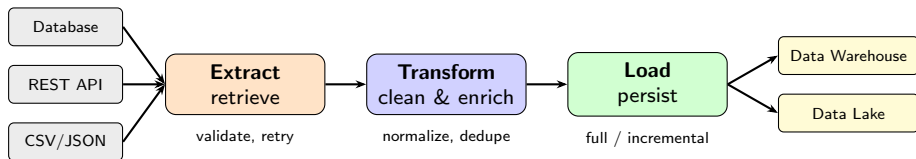
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ETL: Extract, Transform, Load

ETL Pipeline

A **data integration pattern** that **extracts** raw data from heterogeneous sources, **transforms** it to meet quality and schema requirements, and **loads** it into a **destination system** ready for analysis.



- **Extract:** handle **partial failures**, **pagination**, and **auth tokens**
- **Transform:** apply **business rules**, reshape **schemas**, **quality checks**
- **Load:** support **upserts**, **atomicity**, and **rollback** on failure



ETL vs. ELT: Modern Paradigm Shift

ETL (Classic)

Transform **before** loading.
Transforms run on a **dedicated server**.

- **Controlled** schema at load time
- Required with **limited** destination capacity
- Tools: Informatica, Talend

ELT (Modern)

Load raw data **first**; transform inside the warehouse.

- Leverages **cloud-scale** compute
- **Preserves raw** data for replay
- Tools: dbt, Spark, BigQuery

ETL Orchestration Tools

- **Batch**: Apache Airflow (DAG), Luigi, Prefect
- **Stream**: Apache Kafka, Apache Flink, Spark Streaming
- **Cloud-native**: AWS Glue, Azure Data Factory, Google Dataflow

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ETL: Current Challenges

Core Engineering Challenges

Building **reliable**, **scalable**, and **observable** pipelines at production scale is significantly harder than it appears in tutorials.

- **Schema drift**: source systems change columns without notice $\Rightarrow$ pipelines silently break
- **Data volume**: terabyte-scale ingestion requires *partitioning, parallelism, and incremental patterns*
- **Complexity**: ETL pipelines are often composed of *watermarks, stateful operators, and windowing*
- **Performance**: ETL pipelines are often *distributed and parallelized*
- **Reliability**: ETL pipelines are often *distributed and parallelized*



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- **Idempotency**: re-running a failed job must **not** produce duplicate records
- **Observability**: detect failures, measure **data freshness**, track **lineage** across pipeline stages
- **PII compliance**: **GDPR / CCPA** require data **masking**, **deletion**, and **access auditing**



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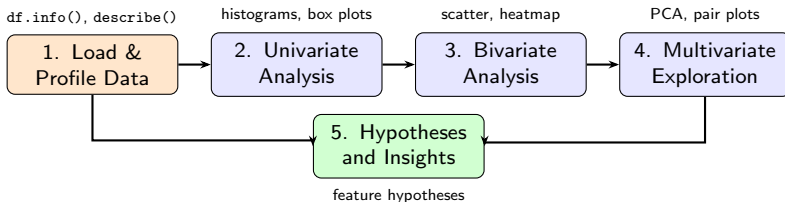
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Exploratory Data Analysis (EDA)

Definition — Tukey (1977)

EDA is the practice of **examining datasets** to summarize their **main characteristics**, discover **patterns**, detect **anomalies**, and **generate hypotheses** before formal modelling.



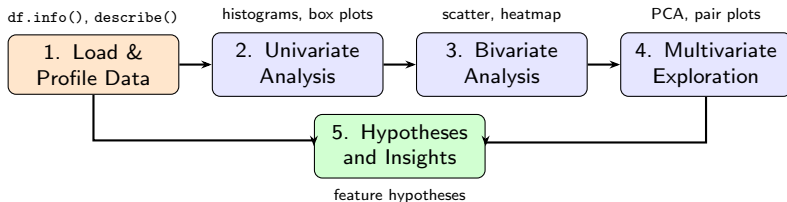
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Descriptive Analytics: Central Tendency and Dispersion

Central Tendency

- **Mean:** $\bar{x} = \frac{1}{n} \sum x_i$ — **sensitive** to outliers
- **Median:** middle value; **robust** to outliers
- **Mode:** most frequent; useful for **categorical** data

Dispersion

- **Variance:**
 $s^2 = \frac{1}{n-1} \sum (x_i - \bar{x})^2$
- **Std Dev:** $s = \sqrt{s^2}$; same units as data
- **IQR:** $Q_3 - Q_1$; **robust** spread measure
- **Range:** $x_{\max} - x_{\min}$; sensitive to extremes

Normal distribution — [≈68]-[≈95]-[≈99.7] rule



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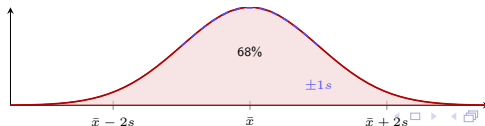
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Quantiles, Skewness, and Kurtosis

Quantiles

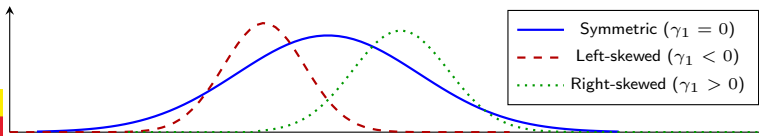
Values that **partition** a sorted dataset into equal groups.

- Q_1, Q_2, Q_3 : **quartiles**
- P_{95} : 95th **percentile** — SLA thresholds
- **IQR** = $Q_3 - Q_1$: robust **spread** measure

Skewness and Kurtosis

- $\gamma_1 > 0$: right-skewed (long right tail)
- $\gamma_1 < 0$: left-skewed (long left tail)
- $\gamma_2 > 0$: leptokurtic — heavy tails
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Distribution shapes



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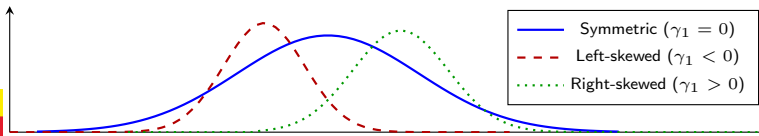
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Correlation and Association in EDA

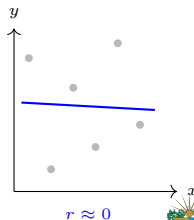
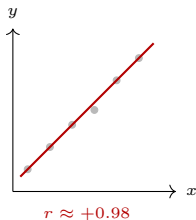
Pearson Correlation r

Measures **linear association** between two continuous variables, $r \in [-1, 1]$.

Correlation does not imply causation.

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \cdot \sum (y_i - \bar{y})^2}}$$

- $|r| \geq 0.7$: **strong**
 $0.4 \leq |r| < 0.7$: **moderate**
- **Spearman ρ** : rank-based,
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 monotonic relationship
- **Kendall τ** : concordant
 pairs; for **small samples** or
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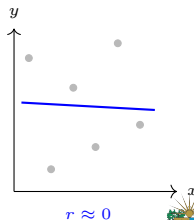
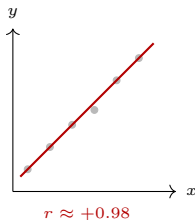
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Correlation does not imply Causation

This keeps happening. How heavy are cats?



EDA Applications: Connecting to the Pipeline

EDA as a Quality Gate

EDA is not just exploratory — it **validates pipeline outputs** at the **Silver layer** before Gold aggregations are computed.

What EDA reveals:

- **Distribution shifts** between ingestion runs
- Unexpected **null rates** in critical columns
- **Duplicate** records from concurrent DAG runs
- Temporal **gaps** — missing ingestion windows
- **Cardinality** anomalies in categorical columns

EDA tools (Python):

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* pandas: .describe(),
      .value_counts(),
      .isnull().sum()
* ydata-profiling: auto full
      report
* seaborn: pairplot(),
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* plotly: interactive
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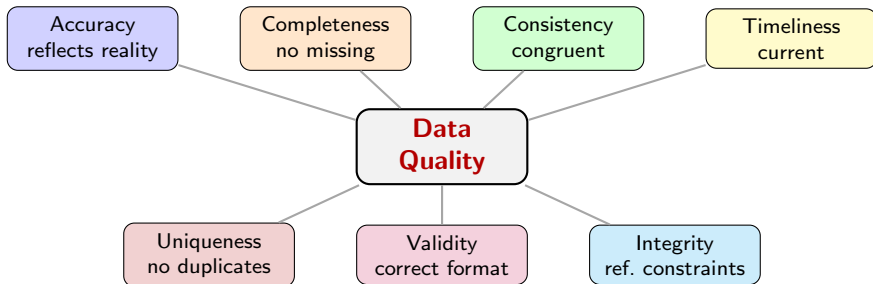
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Data Quality Dimensions

“Garbage In, Garbage Out”

Poor data quality leads to **biased models**, *incorrect decisions*, and **wasted engineering resources**. Quality must be **measured**, not assumed.



Outlier Handling I: *Detection Methods*

Outlier

An observation that **deviates markedly** from other data points. May be a **measurement error**, a **rare genuine event**, or a **data quality issue** — context determines the treatment.

Statistical Methods:

- **Z-score:**
 $|z_i| > 3$; $z_i = (x_i - \mu) / \sigma$
 — assumes normality
- **IQR Rule:**
 $x < Q_1 - 1.5 \cdot \text{IQR}$ or
 $x > Q_3 + 1.5 \cdot \text{IQR}$
 — robust
- **Modified Z-score:** uses median instead of mean; better for skewed data

Algorithmic Methods:

- **Isolation Forest:** random splits; anomalies isolated in fewer partitions
- **DBSCAN:** density-based; points in low-density regions are outliers
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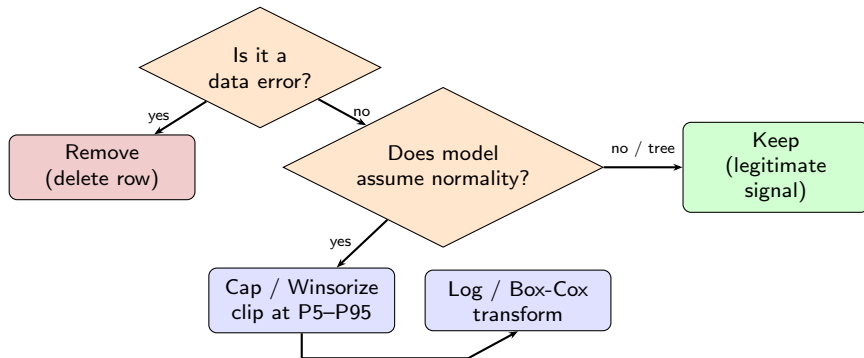
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Outlier Handling II: *Treatment Strategies*



Practical Guidance

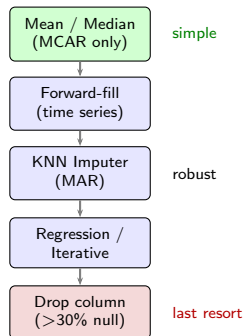
- **Always** investigate outliers before removing: they may encode the most valuable information.
- Tree-based models (RandomForest, XGBoost) are **naturally robust** to outliers — no treatment needed.

Missing Data: Mechanisms and Imputation

Missing Data Mechanisms

- **MCAR:** *Completely at Random* — absence **unrelated** to any variable; safest.
- **MAR:** *At Random* — depends on **observed** variables; imputation possible.
- **MNAR:** *Not at Random* — depends on the **missing value itself**; risk of bias.

Imputation Methods



Rule of Thumb

< 5% missing → mean/median.

5–30% → model-based imputer.

> 30% → **consider dropping** the column.

Feature Engineering

Definition

The process of transforming **raw data** into **informative features** that better expose the **underlying structure** to learning algorithms. Feature engineering often has **more impact** than algorithm choice.



Feature Engineering

Encoding categorical variables:

- **One-Hot:** binary indicator per category; sparse
- **Target encoding:** replace with conditional mean of target
- **Embeddings:** dense learned representation

Numerical transformations:

- **Scaling:** standardize ($\mu = 0, \sigma = 1$) or min-max $[0, 1]$
- **Log / Box-Cox:** reduce right-skewness
- **Polynomial:** x^2, x_1, x_2 to capture non-linearities

Domain-specific features:

- **Temporal:** day-of-week, holiday flag, season, lag features
- **Text (NLP):** TF-IDF, sentiment score, word counts
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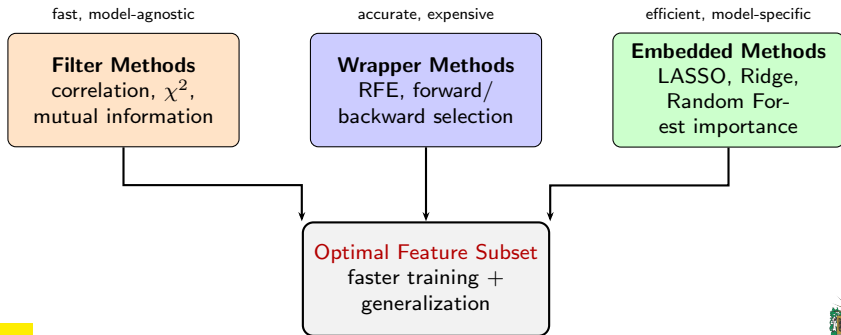
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Feature Selection

Why Select Features?

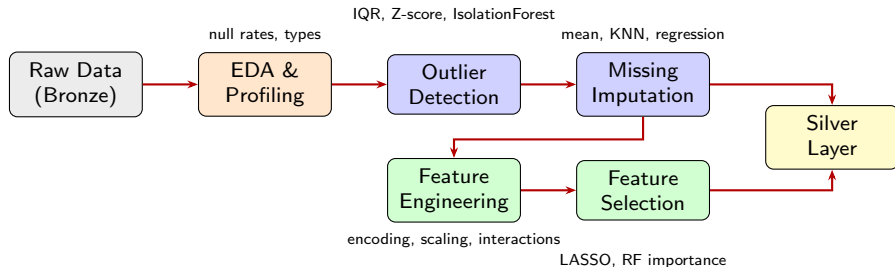
Reducing the **feature space** lowers **overfitting risk**, speeds up training, and improves **model interpretability**. **Irrelevant** or **redundant** features add noise.



Data Curation Pipeline: End to End

Data Curation

The systematic process of **discovering**, **cleaning**, **enriching**, and **organizing** data to make it **fit for purpose** for analysis and machine learning.



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- 

Data Curation: Current Challenges

Why Curation is Hard in Practice

Real-world data violates every textbook assumption simultaneously. **80% of data science work** is data preparation.

- **Concept drift:** the **statistical distribution** of features changes over time — models become *stale*
- **Class imbalance:** rare events (**fraud, churn**) underrepresented — SMOTE, class weights
- **Leakage:** future information **accidentally** included in training features — invalidates evaluation
- **High cardinality:** millions of unique categories (**user IDs**) — hashing, embeddings required
- **Multicollinearity:** highly **correlated features** inflate variance in linear models — VIF analysis, PCA

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Thanks!

Questions?



Repo: <https://github.com/EngAndres/ud-public/tree/main/courses/data-analysis-programming>

